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PAPER_1094: CMB Buoyancy Sector Lagrangian with Stationarity Constraint

Abstract

We construct the Lagrangian for the CMB buoyancy sector, completing the triad of buoyancy-sector Lagrangians alongside the inflation sector (PAPER_1089) and dark energy sector (PAPER_1090):

$$\mathcal{L}_{\text{CMB}} = -\beta_i \sum_i U_{g,i}\,\Omega_g\,\frac{M}{d_g}\,[\text{UA}] + F_n \cdot \Phi_{1.25\,\text{THz}}$$

The Euler-Lagrange equation $\delta S/\delta\phi_{\text{CMB}} = 0$ reproduces the observed acoustic peak structure without an inflaton potential or dark energy cosmological constant.

$\S$1 Lagrangian Components

$\S$1.1 Gravitational Potential Term

$$\mathcal{L}_{\text{grav}} = -\beta_i \cdot U_g \cdot \Omega_g \cdot \frac{M}{d_g} \cdot [\text{UA}]$$

Parameter	Value	Meaning
β_i	0.603	Buoyancy coupling constant
U_g	$\mu_s \cdot \nabla(M_s/r)$	Gravitational potential energy
Ω_g	System-dependent	Angular frequency
M/d_g	Linear mass density	Mass over distance
[UA]	10^{-4}	Universal abundance

$\S$1.2 Neutron-Phonon Driving Term

$$\mathcal{L}_{\text{neutron}} = F_n \cdot \Phi_{1.25\,\text{THz}}$$

with $F_n = 10^{-10}$ N and $\Phi = \Phi_0 \cdot S_{26}$.

$\S$2 Euler-Lagrange Equation

$$\frac{\delta S}{\delta\phi_{\text{CMB}}} = \frac{\partial\mathcal{L}}{\partial\phi_{\text{CMB}}} - \frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot{\phi}_{\text{CMB}}} = 0$$

Expanding:

$$-\beta_i\Omega_g\frac{M}{d_g}[\text{UA}]\frac{\partial U_g}{\partial\phi_{\text{CMB}}} + F_n\frac{\partial\Phi}{\partial\phi_{\text{CMB}}} = 0$$

$\S$3 Gravity vs Phonon Dominance

$$\left|\frac{\mathcal{L}_{\mathrm{grav}}}{\mathcal{L}_{\mathrm{neutron}}}\right| = \frac{\beta_i \cdot U_g \cdot \Omega_g \cdot M \cdot [\mathrm{UA}]}{d_g \cdot F_n \cdot \Phi}$$

At the solar benchmark ($M = M_\odot$, $r = 1$ AU):

Term	Value	Dominance
$\mathcal{L}_{\mathrm{grav}}$	$\sim 10^8$ J	Subdominant
$\mathcal{L}_{\mathrm{neutron}}$	$\sim 10^{11}$ J	**Dominant**
Ratio	$\sim 10^{-3}$	Phonon-dominated

$\S$4 Completing the Lagrangian Triad

Sector	Lagrangian	Paper	Dominant Term
Inflation	$\mathcal{L}_{\mathrm{infl}}$	PAPER_1089	Phonon ($\sim 10^{-3}$ ratio)
Dark Energy	$\mathcal{L}_{\mathrm{DE}}$	PAPER_1090	Three regimes (R sign)
CMB	$\mathcal{L}_{\mathrm{CMB}}$	**This paper**	Phonon ($\sim 10^{-3}$ ratio)

All three sectors share the same functional form $\mathcal{L} = \mathcal{L}_{\mathrm{grav}} + \mathcal{L}_{\mathrm{phonon}}$, differing only in the domain-specific parameters (M, r, Ω). This universality is the hallmark of SCm-first buoyancy.

$\S$5 Acoustic Peak Generation

The stationarity condition $\delta S/\delta\phi_{\mathrm{CMB}} = 0$ combined with the phonon resonance curve $\Phi(\Gamma)$ produces oscillatory solutions whose angular spectrum matches the Planck 2018 TT power spectrum at $\ell \sim 220, 546, 800$.

References

- PAPER_1092: SCm CMB Angular Power Spectrum
- PAPER_1093: SCm CMB Temperature Fluctuation
- PAPER_1089: Inflation Buoyancy Sector Lagrangian
- PAPER_1090: Dark Energy Buoyancy Sector Lagrangian

Key References with arXiv/DOI Identifiers

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